

Notation

Nobody reads a table of symbols and remembers it, and nothing later assumes you did. Four lines are enough to start Lecture 1, slides 1–63: an action is a and a reward is r ; T is how many rounds there are; $\mu(a)$ is the unknown mean payoff of a and $\hat{\mu}(a)$ the estimate of it; and $\text{Reg}(T)$ is what not knowing has cost by round T . The rest arrives when it is needed, and this chapter is here to come back to.

Two symbols cause more trouble than the others, both because the literature is divided about them. T here is the horizon: Bellman operators are $\mathcal{B}$, transposition is $^\top$, and a paper that writes T for either of those means something this book calls by another name. And a cost is something to minimise while an objective is something to maximise, so they get different letters: $\mathcal{L}$ goes down, J goes up.

Core objects

symbol	meaning
$s \in \mathcal{S}, a \in \mathcal{A}$	state, action; sets are calligraphic
s'	successor state
$ \mathcal{S} , \mathcal{A} $	cardinalities; the bars matter, since a bare S would read as a state
t	time step
k	iteration index of an algorithm (value iteration sweeps, policy improvement steps)
T	horizon: bandit rounds, episode length, finite-horizon MDP length
$\gamma \in [0, 1)$	discount factor; effective horizon $1/(1 - \gamma)$
$p(s' s, a)$	transition dynamics
P^π	transition matrix induced by π , when the matrix form is needed
$r(s, a) \in [0, R_{\max}]$	reward
$\mathbf{r}^\pi$	vector of expected rewards under π
$\pi(a s), \pi(s)$	stochastic, deterministic policy
τ	trajectory; $\tau_{0:t}$ is the history up to time t
μ	initial state distribution
d_μ^π	discounted state-action distribution under π from μ

Reward scale. Rewards live in $[0, R_{\max}]$ throughout. The bandit chapters set $R_{\max} = 1$, which is why their regret bounds look exactly like the ones in the bandit literature. Elsewhere $R_{\max}$ stays in the formulas, so a bound can be read at whatever scale a problem actually has.

Values and operators

symbol	meaning
V^π, Q^π, A^π	state value, action value, advantage

symbol	meaning
V^*, Q^*, π^*	optimal value, optimal action value, optimal policy
$\mathcal{B}$	Bellman optimality operator
$\mathcal{B}^\pi$	Bellman expectation operator for π
$\hat{V}, \hat{Q}, \hat{\mu}$	estimates; the hat sits on the quantity being estimated, so the letter still says what it is
G_t	return from time t ; $G_t^{(n)}$ the n -step return, G_t^λ the λ -return
δ_t	temporal-difference error

Part of the literature, Bertsekas and Agarwal–Jiang–Kakade–Sun among others, writes T for the Bellman operator. A reader who follows the references will meet it, and it is the same object as $\mathcal{B}$.

Approximation and learning

symbol	meaning
$\phi(s, a)$	feature vector
w	parameters of a value function; a policy's parameters are θ
θ	parameters of a policy; a value function's parameters are w
α	learning rate
λ	eligibility-trace decay; also the weight of the λ -return
$e_t(s)$	eligibility trace of s at time t
ϵ	exploration rate (ϵ -greedy); tolerances are ϵ
$\varepsilon_{\text{approx}}$	approximation tolerance in a bound
δ_{KL}	trust-region radius
z	a generic observed input: the context of a contextual bandit, the sample point of a regression
$\mathcal{D}$	dataset; $\mathcal{D}_{\text{exp}}$ for expert demonstrations
ρ	distribution the data is drawn from
n	number of samples in a dataset
$N_t(a)$	number of times a has been taken up to t
$\mathcal{L}(w)$	loss, minimised
$J(\pi_\theta)$	policy objective, maximised
η	observation noise on a regression label

The subscripts on $\varepsilon_{\text{approx}}$ and δ_{KL} earn their keep. ε and ϵ are too close to carry a distinction on their own, and a bare δ is the TD error.

Bandits

symbol	meaning
K	number of arms, in the bandit chapters
ν_a	reward distribution of arm a
$\mu(a)$	its mean; $\mu(a^*)$ is the best arm's mean
β	confidence multiplier in a UCB bonus
δ	failure probability of a confidence bound, so bounds read as $1 - \delta$; elsewhere in the book δ is the TD error
$\text{Reg}(T)$	regret over T rounds
A_t	the arm pulled at round t ; some texts write I_t
a^*	an optimal arm, $a^* \in \arg \max_a \mu(a)$
$\hat{\mu}_t(a)$	empirical mean of arm a from the pulls before round t
m	pulls per arm during the exploration phase of Explore-then-Commit
$\text{rad}(n, \delta)$	Hoeffding radius, $\sqrt{\ln(2/\delta)/(2n)}$, the constant lab L1 checks

Optimal control

Vectors are lowercase bold, matrices uppercase bold.

symbol	meaning
x_t, u_t	state and control input
f	dynamics in general, $x_{t+1} = f(x_t, u_t)$, linear or not
c, c_H	running cost and terminal cost
$\mathbf{F}_x, \mathbf{F}_u$	linear dynamics: $x_{t+1} = \mathbf{F}_x x_t + \mathbf{F}_u u_t$
$\mathbf{C}_x, \mathbf{C}_u$	quadratic cost: $c(x, u) = x^\top \mathbf{C}_x x + u^\top \mathbf{C}_u u$
$\mathbf{P}_t$	Riccati matrix, $V_t^*(x) = x^\top \mathbf{P}_t x$
$\mathbf{K}_t$	feedback gain, $u = \mathbf{K}_t x$

Why not Q and R . Every control textbook writes the cost matrices as Q and R . Here they are $\mathbf{C}_x$ and $\mathbf{C}_u$, for the sake of the line that results when the two traditions meet on one page:

$$Q_t^*(x, u) = x^\top Q x + u^\top R u + \dots$$

where Q on the left is an action-value function and Q on the right is a cost matrix. After six chapters spent building Q as *the value of a choice*, that line cannot be defended, and no typographic convention rescues it. $\mathbf{C}_x$ and $\mathbf{C}_u$ read as what they are. When you go to the control literature, $\mathbf{C}_x$ is its Q and $\mathbf{C}_u$ is its R .

Mathematical furniture

symbol	meaning
$\top$	transpose; T is the horizon, so it cannot be this
$*$	optimal
$\mathbf{1}[\cdot]$	indicator function
$\mathbf{I}$	identity matrix
$\mathbb{E}, \mathbb{P}$	expectation, probability
$\mathbb{R}$	the reals
$\mathcal{N}(m, \sigma^2)$	Gaussian with mean m and variance σ^2
$D_{\text{KL}}(p \parallel q)$	Kullback–Leibler divergence
$\mathcal{H}(\pi)$	entropy of a policy
$\ \cdot\ _\infty$	sup-norm, the norm every contraction argument uses